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In [1]: import pandas as pd
import numpy as np
import joblib

from sklearn.preprocessing import StandardScaler, LabelEncoder, OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer

from xgboost import XGBClassifier

RANDOM_STATE = 87941

In [2]: # Load data
df = pd.read_csv("../data/Crop_Yield_Fertilizer.csv")

In [3]: NUM_FEATURES = ['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall']
CAT_FEATURES = ['label']
TARGET = 'fertilizer'

X = df[NUM_FEATURES + CAT_FEATURES]
y = df[TARGET]

In [4]: fertilizer_encoder = LabelEncoder()
y_encoded = fertilizer_encoder.fit_transform(y)

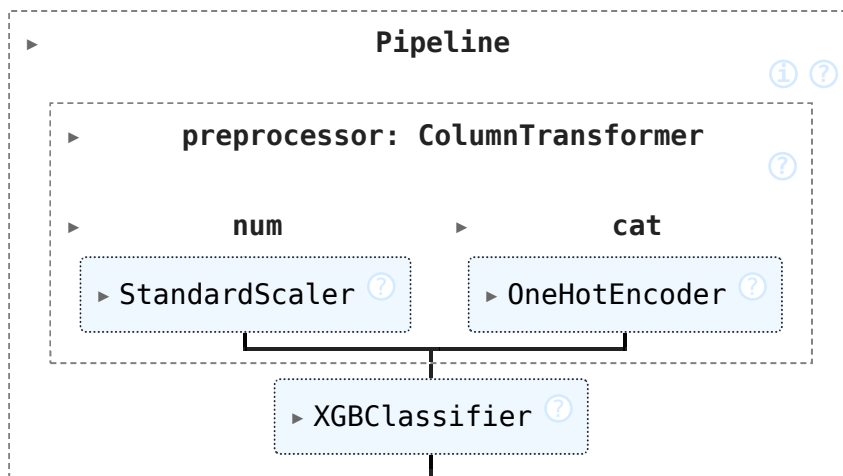
In [5]: preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), NUM_FEATURES),
        ('cat', OneHotEncoder(handle_unknown='ignore', sparse_output=False), CAT_FEATURES)
    ]
)

In [6]: model = XGBClassifier(
    n_estimators=600,
    max_depth=6,
    learning_rate=0.05,
    subsample=0.85,
    colsample_bytree=0.85,
    objective='multi:softprob',
    num_class=len(fertilizer_encoder.classes_),
    eval_metric='mlogloss',
    random_state=RANDOM_STATE
    # n_jobs=-1
)

In [7]: pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('model', model)
])

In [8]: pipeline.fit(X, y_encoded)
```

Out[8]:



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In [10]: joblib.dump(pipeline, "../models/fertilizer_model.pkl")
         joblib.dump(fertilizer_encoder, "../models/fertilizer_label_encoder.pkl")

         print("Fertilizer model trained and saved")
```

Fertilizer model trained and saved